

Joshua Greenhalgh

SENIOR BACKEND SOFTWARE ENGINEER

United Kingdom

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Summary

Backend software engineer working in Python and PostgreSQL/PostGIS since 2017, in senior roles across biotech, adtech, climate analytics and humanitarian data, including two years at the non-profit Flowminder Foundation. Builds the backend systems that ingest, validate and transform large partner and customer datasets: geospatial workflows on PostGIS, satellite imagery pipelines feeding vision models, data checks, error handling and observability, and Terraform-managed containerised services with CI/CD on AWS and GCP. Traces problems through data and system behaviour, is comfortable with ambiguous and evolving requirements, mentors engineers and explains complex ideas clearly to non-technical colleagues, backed by a first-class mathematics degree and doctoral research in computational modelling.

Professional Experience

Independent Software & ML Projects

UK

OPEN-SOURCE TOOLING AND EXPERIMENTS

May 2026 – Present

- Building and open-sourcing ML tooling: an activation-probing toolkit and a weight-perturbation experiment framework for open language models, a Modal and Weights & Biases experiment runner, and an LLM-based knowledge-graph extraction pipeline (see Projects).

Senior Software Engineer

UK

IMU BIOSCIENCES

Feb 2024 – May 2026

- Designed and built the **data platform and catalogue** for 100k+ single-cell (flow cytometry) samples, unifying scattered Parquet and vendor file sources into a structured, lineage-aware index. Cut query times and made ML training datasets fully reproducible.
- Owned **storage architecture** for high-volume single-cell data end to end: benchmarked Zarr and Parquet access patterns across S3 and Snowflake and delivered a hybrid design serving both whole-sample retrieval and cross-sample aggregation for model training.
- Built a **QC and monitoring framework** in Dagster that automatically validates reference-sample consistency across experiments, catching data-quality regressions before they reach analysts or models.
- Defined **domain ontologies and metadata standards** with immunologists and embedded them in pipeline outputs, standardising labelling across diverse assays and improving the interpretability of downstream ML models.
- Developed a **synthetic data generation framework** that produces realistic single-cell datasets, used to stress-test pipelines and validate ML model generalisation.

CAREER BREAK: PLANNED TIME OFF BETWEEN ROLES, WALKED THE CAMINO DE SANTIAGO

Oct 2023 – Jan 2024

Senior Data Engineer

UK

MOBKOI

Jan 2022 – Oct 2023

- Rebuilt the company's data pipelines from **ad-hoc cloud functions to Prefect orchestration** on GCP with CI/CD, increasing deployment speed and safety while reducing operational overhead.
- Led migration from aggregated partner data to **impression-level ingestion**, integrating with AWS ad-serving infrastructure and CloudWatch logs, enabling granular analytics and increasing client confidence in reported numbers.
- Designed and implemented a **new data warehouse** with change data capture from internal operational systems, improving data freshness and reliability for the analytics team.
- Overhauled **revenue estimation and FX handling**, reducing monthly variance against P&L from ~20% to under 1%.

Co-founder

UK

PERSONAL LOCATION DATA UNION (CO-FOUNDED WITH A FORMER FLOWMINDER COLLEAGUE)

Apr 2021 – Dec 2021

- Two-sided data venture: individuals share their location data on their own terms and are paid for it, and buyers receive **privacy-preserving mobility aggregates**, building on Flowminder experience of turning raw location data into aggregate mobility products.
- Built a proof-of-concept **mobile app (Expo/React Native) and location collection backend on GCP** (BigQuery, Terraform) end to end, from consent and data capture through storage and aggregation. Discontinued after finding the pay-for-data model commercially unviable.

Senior Data Engineer

UK

TRANSITIONZERO

Jan 2021 – Mar 2021

- Implemented **Terraform-managed infrastructure on GCP** with revision-based deployments, and replaced Cloud Composer with **Prefect pipelines** and a CI-based framework that let analysts deploy their own pipelines independently.
- Built satellite imagery processing pipelines on **Google Earth Engine** feeding vision-based ML models for climate analytics.

Senior Data Engineer

HEYJOBS

Berlin, Germany

Aug 2019 – Nov 2020

- Built a **data quality framework for the Redshift warehouse** using property-based testing of SQL transformations, integrated with Airflow, DataDog and Slack as part of CI. Significantly reduced post-deployment data issues.
- **Tech lead** for redesigning marketing cost processing, migrating undocumented legacy SQL to fully unit-tested Python and reducing marketing overspend by ~5%.
- Established **Airflow best practices** (incremental, idempotent pipelines) and moved intermediary data to Glue/Spectrum, cutting Redshift costs.

Data Engineer / Analyst

FLOWMINDER FOUNDATION

Southampton, UK & Berlin (remote)

Jun 2017 – Jun 2019

- **Led a GSMA-funded project end to end**, including budget and contractors, standardising ETL on **Airflow** across multiple countries (50M+ mobile phone records ingested daily) and building a post-crisis mobility analysis tool on Kafka and TimescaleDB for humanitarian aid targeting.
- Led an **Inter-American Development Bank** project estimating commuting flows in Haiti's two largest cities from three years of mobile phone data (~10M subscribers), delivering interactive PostGIS and React dashboards to the transport ministry.
- Designed and delivered a two-week **Python and R training programme** for Nepal's National Statistics Office on JupyterHub (GKE).

Doctoral Researcher, Computational Modelling

UNIVERSITY OF SOUTHAMPTON

Southampton, UK

Sep 2015 – Jun 2017

- Researched **deep learning, optimisation and Bayesian methods** for inverse problems in x-ray tomography (TensorFlow, PyMC3, OpenMP/MPI on HPC). Taught and mentored undergraduates in Python and mathematics and postgraduates in research software engineering. Left to accept a permanent offer from Flowminder after an internship there.

Projects

SELECTED PERSONAL ML & LLM PROJECTS, OPEN SOURCE AT GITHUB.COM/JOSH-GREE

2025 – 2026

- **probe-lab**: toolkit for LLM activation probing. Extracts per-layer residual-stream activations with PyTorch and Hugging Face Transformers and fits linear probes against prompt metadata for interpretability experiments.
- **weight-corruption**: experimental framework measuring how weight perturbations (noise, zeroing, shuffling, quantisation) degrade language modelling and behaviour across the Pythia model family, with GPU evaluation on Modal.
- **birdclef-2026** and **wm**: bird-audio classification in PyTorch (timm, ONNX embeddings) trained on Modal GPUs with Weights & Biases tracking; wrote **wm**, a small framework that snapshots every experiment run as a PR for reproducibility.
- **news-knowledge-graph**: LLM extraction pipeline turning news articles into a knowledge graph using DSPy programs and GLiNER2 for entity extraction, temporal tagging and Wikidata entity linking, orchestrated with Prefect.
- **mobile-agents**: GitHub Actions workflow built on the Claude Agent SDK that plans and implements issues autonomously and opens pull requests.

Skills

Languages & Backend	Python, SQL, Bash, Flask, REST APIs, PySpark, R, JavaScript
PostgreSQL	PostgreSQL, PostGIS, TimescaleDB, data modelling, joins, query optimisation, performance tuning
Geospatial	PostGIS, Google Earth Engine, satellite imagery pipelines, mobility aggregates, geospatial datasets
Cloud & Infrastructure	AWS (S3, Lambda, Glue, EMR, CloudWatch), GCP (BigQuery, GKE, Cloud Functions), Terraform
Containers & Delivery	Docker, Kubernetes, Linux, containerised services, GitHub, GitHub Actions, CI/CD, Agile
Data Pipelines	Dagster, Prefect, Airflow, dbt, Kafka, ETL/ELT, ingestion, validation and transformation, change data capture
Quality & Observability	pytest, property-based testing, data checks, error handling, data quality monitoring, DataDog
Storage & Warehousing	Snowflake, Redshift, Parquet, Zarr, S3, lakehouse
ML & AI	PyTorch, HF Transformers, TensorFlow/Keras, vision models, DSPy, Claude Agent SDK, Weights & Biases

Education

MSc Computational Modelling

UNIVERSITY OF SOUTHAMPTON

Southampton, UK

2014 – 2015

PGCE Mathematics and Qualified Teacher Status

MANCHESTER METROPOLITAN UNIVERSITY

Manchester, UK

2012 – 2014

BSc Mathematics, First Class Honours

MANCHESTER METROPOLITAN UNIVERSITY

Manchester, UK

2009 – 2012